

1. Questions

The following question has three statements as statement I, statement II, and statement III. You have to determine whether the data provided in three statements is sufficient to answer the question or not.

Find the rate of interest offered in scheme A.

Statement I: The simple interest received from Scheme A after 3 years is Rs. 10800.

Statement II: The compound interest received from scheme B at the rate of 10% per annum compounded annually after 2 years is Rs. 4200.

Statement III: The ratio of the amount invested in schemes A and B is 3:2.

- a. Both statements II and III together are necessary
- b. Only statement III alone is sufficient
- c. Both statements I and II together are necessary
- d. All statements I, II, and III together are necessary
- e. Neither statement I or II nor III is sufficient

2. Questions

Find the initial mixture (milk + water) in the vessel.

Statement I: The ratio of the quantity of milk and water in the initial mixture of the vessel is 2:1.

Statement II: If 60 ml is taken out from the initial mixture and replaced with the same quantity of water, then the quantity of milk and water in the resultant mixture will be equal.

Statement III: If 120 ml of water is added to the initial mixture, then the quantity of milk in the resultant mixture will be 40 ml less than water.

- a. Both statements II and III together are necessary
- b. Only statement III alone is sufficient
- c. Either statement II or III along with statement I is necessary
- d. All statements I, II, and III together are necessary
- e. Neither statement I or II nor III is sufficient

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3. Questions

Find the cost price of the article.

Statement I: The article has been sold after allowing a discount of 11.11%.

Statement II: The profit earned on selling the article is Rs. 40.

Statement III: The article has been marked up by 50% above its cost price.

- a. Both statements II and III together are necessary
- b. Only statement III alone is sufficient
- c. Either statement II or III along with statement I is necessary
- d. All statements I, II, and III together are necessary
- e. Neither statement I or II nor III is sufficient

4. Questions

Find the length of the platform.

Statement I: The time taken by the train of length 280 meters to cross the platform is 14 seconds.

Statement II: The train can cross a man running at the speed of 10 m/sec in the opposite direction in 7 seconds.

Statement III: The train can cross a tunnel of length 80 meters in 12 seconds.

- a. Both statements II and III together are necessary
- b. Only statement III alone is sufficient
- c. Either statement II or III along with statement I is necessary
- d. All statements I, II, and III together are necessary
- e. Neither statement I or II nor III is sufficient

5. Questions

Find the time taken by B to complete the whole work alone.

Statement I: A and C together can complete the whole work in $(15/2)$ days.

Statement II: The average of the efficiency of A and B is 2.50.

Statement III: B and C together can complete the whole work in $(20/3)$ days.

- a. Both statements II and III together are necessary
- b. Only statement III alone is sufficient
- c. Both statements I and II together are necessary
- d. All statements I, II, and III together are necessary
- e. Neither statement I or II nor III is sufficient

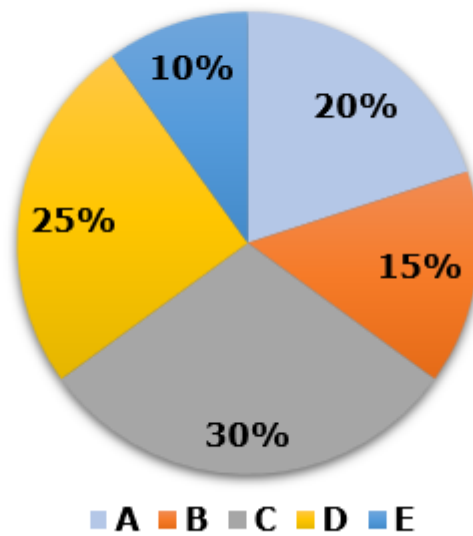
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6. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the percentage distribution of the total number of students who participated in football from five different schools A, B, C, D and E.

Percentage of the total number of students who participated in football



Notes:

- i). The total number of students who participated = the number of students who participated in cricket + the number of students who participated in football + the number of students who participated in hockey
- ii). From each school, the number of students who participated in football is more than in hockey.
- iii). The total number of students who participated (cricket + football + hockey) from schools A, B, C, D, and E is 300, 250, 450, 275, and 150, respectively.
- iv). The difference between the number of students who participated in football and hockey from schools A, B, C, D, and E is 20, 50, 50, 75, and 20, respectively.
- v). The average number of students who participated in cricket from schools A and B is 135.

In cricket, if the ratio of the number of boys who participated from school C to the number of girls who participated from school D is 24:5, while the number of boys who participated from school D is 5 less than the number of girls who participated from school C, then find the total number of boys who participated in cricket from schools C and D together.

- a. 195
- b. 105
- c. 150
- d. 135
- e. None of these

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7. Questions

From school A, the average weight of the students who participated in ice and field hockey is 40 kg and 48 kg, respectively. If the total weight of all students who participated in hockey (ice + field) from school A is 3680 kg, then find the difference between the number of students who participated in ice and field hockey from school A.(From school A, the students participated in ice and

fieldhockey)

- a. 24
- b. 16
- c. 40
- d. Cannot be determined
- e. None of these

8. Questions

If the number of students who participated in football from another school F is the same as school E, while the total number of students who participated from school F is 200, out of which 20% participated in hockey, then find the number of students who participated in cricket from school F.

- a. 75
- b. 60
- c. 140
- d. 110
- e. None of these

9. Questions

Determine the correct relation between [X], [Y], and [Z].

X = The total number of students who participated in football and hockey from school B.

Y = The difference between the number of students who participated in cricket from schools C and E.

Z = The average number of students who participated in hockey from schools A and C.

- a. $X > Y > Z$
- b. $X < Y < Z$
- c. $X > Y < Z$
- d. $X < Y > Z$
- e. $X = Y = Z$

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10. Questions

Which of the following statement(s) is/are definitely true?

Statement I: The ratio of the number of students who participated in cricket and hockey from school E is 5:4.

Statement II: Out of the total number of students who participated in school C, 33.33% participated in football.

- a. Only statement I
- b. Only statement II
- c. Either statement I or II
- d. Both statements I and II
- e. Neither statement I nor II

11. Questions

Two missing series are given below. Both series follow the same pattern.

Series I: 11, 15, 9, 17, __[X]__, 19

Series II: __[X-6]__, ____, ____, ____, ____, __[Y]__

Which of the following is/are definitely true?

I: Both [X] and [Y] are consecutive natural numbers.

II: [Y] is a perfect cube.

III: [X] is an odd integer.

- a. Only III
- b. Both I and III
- c. Only II
- d. All I, II and III
- e. None is true

12. Questions

Two series are given below. Series I has one wrong term, and series II starts with the correct term that will replace the wrong term of series I.

Series I: 40, 10, 80, 20, 180, 40

Series II: __[A]__, 40, 320, 80, 640, __[B]__

Find the correct relation between [A] and [B].

- a. $A > B$
- b. $A < B$
- c. $A \geq B$
- d. $A \leq B$
- e. $A = B$

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13. Questions

Two wrong series are given below. Each series follows a different pattern.

Series I: 24, 13, 15, 24.50, 55, 142.50

Series II: 30, 41, 54, 71, 91, 113

If [m] and [n] are the wrong terms of series I and II respectively, then find the value of $[(n+7)/2m]$.

- a. 11
- b. 2
- c. 5
- d. 14
- e. 4

14. Questions

A series is given below, which has one missing term.

Series: [P], [Q], [R], __[?]__, 292, 3417

P, Q, and R are single-digit natural numbers.

P is a composite number.

The sum of P and Q is equal to R.

R is a perfect square of an odd prime number.

Q is greater than P.

What should come in place of the question mark?

- a. 49
- b. 25
- c. 36
- d. 64
- e. 40

15. Questions

Find the pattern of the series given below.

Series: 23, 20, 26, 14, 38, -10

If the same pattern is followed in another series whose first term is equal to the square of the smallest composite number, then find the sixth term of another series.

- a. 17
- b. 8

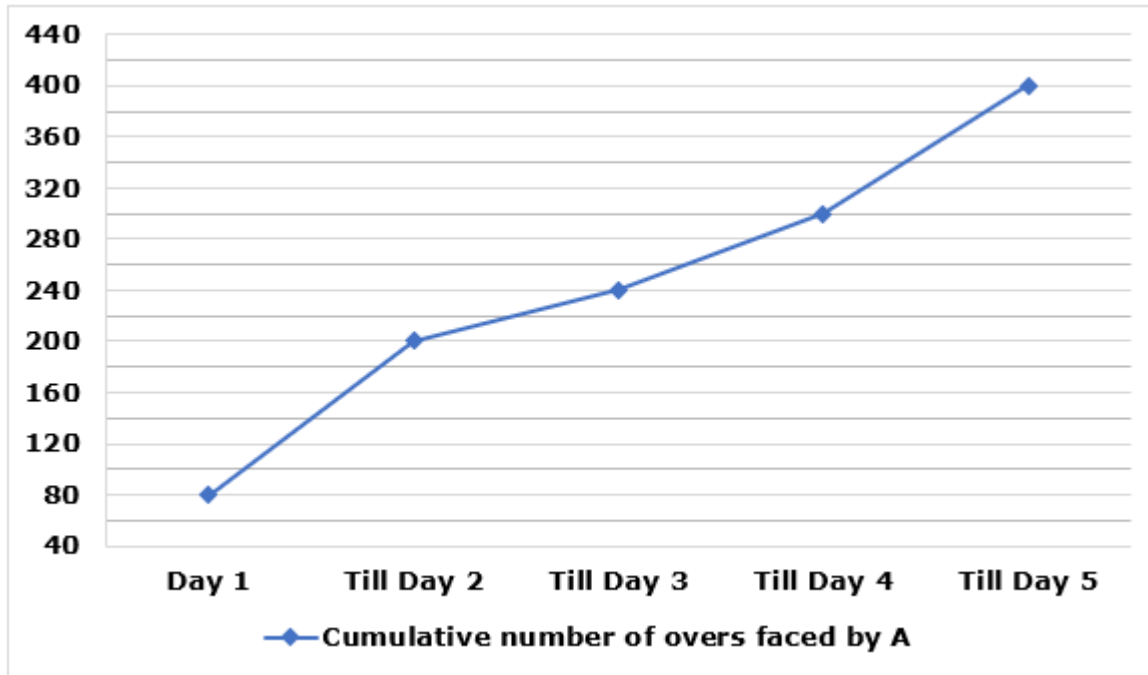
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- c. -8
- d. -17
- e. 11

16. Questions

Read the following information carefully and answer the questions based on it.

The line graph given below shows the cumulative number of overs faced by A from day 1 till day 5 of a test match.



Notes:

- i). 1 over is equal to 6 balls.
- ii). The strike-rate = $\left[\frac{\text{(the number of runs scored)}}{\text{(the number of balls faced)}} \times 100 \right]$
- iii). The ratio of the number of runs scored by A on day 1, day 2, day 3, day 4, and day 5 is 30:48:20:45:60.
- iv). The difference between the strike rate on days 4 and 5 is 30. (day 4 > day 5)

On day 1, if the ratio of the number of fours and sixes scored by A is 6:1 and the number of runs scored in fours is 16% of the number of runs scored other than boundaries by A, then find the total number of runs scored in boundaries (fours + sixes) on day 1 by A.

- a. 42
- b. 88
- c. 60
- d. Cannot be determined
- e. None of these

17. Questions

On day 2, out of the total number of balls faced by A, the number of dot balls is $[2X]$. If the number of dot balls faced is excluded, then the strike rate on day 2 will be increased by 20%. Find the value of $[\sqrt{(X-11)}]$.

- a. 5
- b. 6
- c. 4
- d. 7
- e. None of these

18. Questions

If the total number of runs scored by A on day 3 is increased by $___[?]\%$, but his strike rate remains unchanged, then the number of overs faced by A on day 3 will be 50.

What should come in place of the question mark?

- a. 50%
- b. 25%
- c. 40%
- d. 15%
- e. None of these

19. Questions

If there are six days in a test match and the average number of overs faced by A on all 6 days together is 90, while the strike rate on day 6 is 50, then find the run rate on day 6. [The run-rate = $\{(\text{the number of runs scored})/(\text{the number of overs bowled})\}$]

- a. 3
- b. 2
- c. 4
- d. 5
- e. None of these

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20. Questions

If $\frac{1}{9}$ th of the total number of runs scored by A on day 4 is scored in boundaries and the number of runs scored by A other than boundaries on day 5 is 150 more than the previous day, then find what percent of the total number of runs scored by A on day 5 are scored in boundaries?

- a. 16.67%

- b. 12.50%
- c. 11.11%
- d. Cannot be determined
- e. None of these

Explanations:

1. Questions

Answer: D

Using statements I, II, and III together

The rate of interest offered in scheme B = $[(10+10)+\{(10*10)/100\}]\% = (20+1)\% = 21\%$

So, the amount invested in scheme B = $\{(4200/21)*100\} = \text{Rs. } 20000$

Now, the amount invested in scheme A = $\{(20000/2)*3\} = \text{Rs. } 30000$

Let the rate of interest offered in scheme A be $R\%$ per annum.

So, the simple interest received from Scheme A after 3 years = $(R*3)\%$ of 30000 = $3R\%$ of 30000 = Rs. 900R

Now,

$$900R = 10800$$

$$R = 12$$

Therefore, the rate of interest offered in scheme A = $R\% = 12\%$.

So, all statements I, II, and III together are necessary.

Hence, the correct answer is option D.

2. Questions

Answer: C

Using statements I and II together

Let the initial quantity of milk and water be $2x$ ml and x ml respectively.

So, the total initial mixture = $(2x+x) = 3x$ ml

Resultant Mixture

The total quantity of mixture = $(3x-60)$ ml

The quantity of milk = $2/3$ of $(3x-60) = (2x-40)$ ml

The quantity of water = $[\{1/3 \text{ of } (3x-60)\}+60] = \{(x-20)+60\} = (x+40)$ ml

Now,

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$$2x - 40 = x + 40$$

$$x = 80$$

So, the initial mixture (milk + water) in the vessel = $3x = (3 \times 80) = 240$ ml

Therefore, statements I and II together are necessary.

Using statements I and III together

Let the initial quantity of milk and water be $2y$ ml and y ml respectively.

So, the total initial mixture = $(2y+y) = 3y$ ml

Resultant mixture

The quantity of milk = $2y$ ml

The quantity of water = $(y+120)$ ml

Now,

$$(y+120) - 2y = 40$$

$$y + 80 = 2y$$

$$y = 80$$

So, the initial mixture (milk + water) in the vessel = $3y = (3 \times 80) = 240$ ml

Therefore, statements I and III together are necessary.

So, either statement II or III along with statement I is necessary.

Hence, the correct answer is option C.

3. Questions

Answer: D

Using statements I, II, and III together

The ratio of the marked price and selling price = $9:8$. Let it be Rs. $9n$ and Rs. $8n$ respectively. (**Note:** $11.11\% = 1/9$)

The C.P. = Rs. $(8n-40)$

Now,

$$(8n-40)/9n = 100/150$$

$$(8n-40)/9n = 2/3$$

$$24n - 120 = 18n$$

$$6n = 120$$

$$n = 20$$

So, the cost price of the article = $(8n-40) = (8 \times 20 - 40) = (160 - 40) = \text{Rs. } 120$.

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Therefore, all statements I, II, and III together are necessary.

Hence, the correct answer is option D.

4. Questions

Answer: C

Using statements I and II together

Let the speed of the train be V km/hr.

Now, when the train is crossing the man,

$$\{280/(V+10)\} = 7$$

$$7V + 70 = 280$$

$$7V = 210$$

$$V = 30$$

So, the speed of the train = $V = 30$ km/hr

Let the length of the platform be L meters.

Now, when the train is crossing the platform,

$$\{(280+L)/30\} = 14$$

$$280 + L = 420$$

$$L = 140$$

So, the length of the platform = $L = 140$ meters.

Therefore, statements I and II together are necessary.

Using statements I and III together

Let the speed of the train be S km/hr.

When the train crosses the tunnel,

$$\{(280+80)/S\} = 12$$

$$12S = 360$$

$$S = 30$$

Let the length of the platform be T meters

Now, when the train is crossing the platform,

$$\{(280+T)/30\} = 14$$

$$280 + T = 420$$

$$T = 140$$

So, the length of the platform = $T = 140$ meters.

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Therefore, statements I and III together are necessary.

So, either statement II or III along with statement I is necessary.

Hence, the correct answer is option C.

5. Questions

Answer: D

Using statements I, II, and III together

The time taken to complete the whole work by,

$$(A+C) = (15/2) \text{ days}$$

$$(B+C) = (20/3) \text{ days}$$

The total work = 60 units
(L.C.M. of the time taken)

So, the efficiency of,

$$(A+C) = \{60/(15/2)\} = 8$$

$$(B+C) = \{60/(20/3)\} = 9$$

$$B - A = 1 \dots\dots\dots(i)$$

$$B + A = (2.50 \times 2)$$

$$B + A = 5 \dots\dots\dots(ii)$$

By solving equation (i) and (ii), we get,

$$2A = 4$$

$$A = 2$$

Now, put the value of A in equation (i), we get,

$$B - A = 1$$

$$B - 2 = 1$$

$$B = 3$$

Therefore, the time taken by B to complete the whole work alone = $(60/3) = 20$ days.

So, all statements I, II, and III together are necessary.

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Hence, the correct answer is option D.

6. Questions

Let the total number of students who participated in football from all five schools together be 100k.

Now,

The total number of students who participated from school A = 300

The number of students who participated in football from school A = 20% of 100k = 20k

So, the number of students who participated in hockey from school A = $(20k-20)$

And, the number of students who participated in cricket from school A = $\{300-(20k+20k-20)\} = (320-40k)$

Also,

The total number of students who participated from school B = 250

The number of students who participated in football from school B = 15% of 100k = 15k

So, the number of students who participated in hockey from school B = $(15k-50)$

And, the number of students who participated in cricket from school B = $\{250-(15k+15k-50)\} = (300-30k)$

Now,

$$(320-40k) + (300-30k) = (135*2)$$

$$620 - 70k = 270$$

$$70k = 350$$

$$k = 5$$

Therefore, the total number of students who participated from all five schools together = $100k = (100*5) = 500$.

School A

The total number of students who participated = 300

The number of students who participated in football = 20% of 500 = 100

The number of students who participated in hockey = $(100-20) = 80$

The number of students who participated in cricket = $\{300-(100+80)\} = (300-180) = 120$

School B

The total number of students who participated = 250

The number of students who participated in football = 15% of 500 = 75

The number of students who participated in hockey = $(75-50) = 25$

The number of students who participated in cricket = $\{250-(75+25)\} = (250-100) = 150$

School C

The total number of students who participated = 450

The number of students who participated in football = 30% of 500 = 150

The number of students who participated in hockey = $(150-50) = 100$

The number of students who participated in cricket = $\{450-(150+100)\} = (450-250) = 200$

School D

The total number of students who participated = 275

The number of students who participated in football = 25% of 500 = 125

The number of students who participated in hockey = (125-75) = 50

The number of students who participated in cricket = {275-(125+50)} = (275-175) = 100

School E

The total number of students who participated = 150

The number of students who participated in football = 10% of 500 = 50

The number of students who participated in hockey = (50-20) = 30

The number of students who participated in cricket = {150-(50+30)} = (150-80) = 70

School	The total number of students who participated	The number of students who participated in cricket	The number of students who participated in football	The number of students who participated in hockey
A	300	120	100	80
B	250	150	75	25
C	450	200	150	100
D	275	100	125	50
E	150	70	50	30
Total	1425	640	500	285

Answer: A

Now, according to the question,

The number of students who participated in cricket from school C = 200

The number of students who participated in cricket from school D = 100

Let the number of boys who participated from school C and girls who participated from school D be 24m and 5m respectively.

So, the number of girls who participated from school C = (200-24m)

And, the number of boys who participated from school D = (100-5m)

Now,

$$(200-24m) - (100-5m) = 5$$

$$195 - 24m = 100 - 5m$$

$$19m = 95$$

$$m = 5$$

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Therefore, the total number of boys who participated in cricket from schools C and D together = $\{24m + (100 - 5m)\} = (19m + 100) = (19 \times 5 + 100) = (95 + 100) = 195$.

Hence, the correct answer is option A.

7. Questions

Let the total number of students who participated in football from all five schools together be 100k.

Now,

The total number of students who participated from school A = 300

The number of students who participated in football from school A = 20% of 100k = 20k

So, the number of students who participated in hockey from school A = (20k - 20)

And, the number of students who participated in cricket from school A = $\{300 - (20k + 20k - 20)\} = (320 - 40k)$

Also,

The total number of students who participated from school B = 250

The number of students who participated in football from school B = 15% of 100k = 15k

So, the number of students who participated in hockey from school B = (15k - 50)

And, the number of students who participated in cricket from school B = $\{250 - (15k + 15k - 50)\} = (300 - 30k)$

Now,

$$(320 - 40k) + (300 - 30k) = (135 \times 2)$$

$$620 - 70k = 270$$

$$70k = 350$$

$$k = 5$$

Therefore, the total number of students who participated from all five schools together = $100k = (100 \times 5) = 500$.

School A

The total number of students who participated = 300

The number of students who participated in football = 20% of 500 = 100

The number of students who participated in hockey = $(100 - 20) = 80$

The number of students who participated in cricket = $\{300 - (100 + 80)\} = (300 - 180) = 120$

School B

The total number of students who participated = 250

The number of students who participated in football = 15% of 500 = 75

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The number of students who participated in hockey = $(75-50) = 25$

The number of students who participated in cricket = $\{250-(75+25)\} = (250-100) = 150$

School C

The total number of students who participated = 450

The number of students who participated in football = 30% of 500 = 150

The number of students who participated in hockey = $(150-50) = 100$

The number of students who participated in cricket = $\{450-(150+100)\} = (450-250) = 200$

School D

The total number of students who participated = 275

The number of students who participated in football = 25% of 500 = 125

The number of students who participated in hockey = $(125-75) = 50$

The number of students who participated in cricket = $\{275-(125+50)\} = (275-175) = 100$

School E

The total number of students who participated = 150

The number of students who participated in football = 10% of 500 = 50

The number of students who participated in hockey = $(50-20) = 30$

The number of students who participated in cricket = $\{150-(50+30)\} = (150-80) = 70$

School	The total number of students who participated	The number of students who participated in cricket	The number of students who participated in football	The number of students who participated in hockey
A	300	120	100	80
B	250	150	75	25
C	450	200	150	100
D	275	100	125	50
E	150	70	50	30
Total	1425	640	500	285

Answer: C

School A

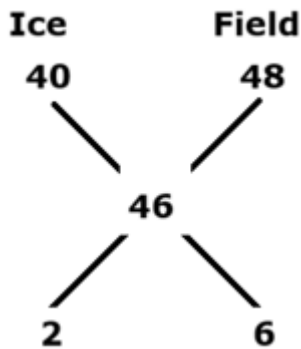
The average weight of those students who participated in ice hockey = 40 kg

And, the average weight of those students who participated in field hockey = 48 kg

The total number of students who participated in hockey (ice + field) = 80

So, the average weight of all students who participated in ice hockey (ice + field) = $(3680/80) = 46$ kg

Now, using the allegation method,



So, the ratio of the number of students who participated in ice and field hockey = $2:6 = 1:3$.

Therefore, the required difference = $2/4$ of $80 = 1/2$ of $80 = 40$.

Hence, the correct answer is option C.

8. Questions

Let the total number of students who participated in football from all five schools together be $100k$.

Now,

The total number of students who participated from school A = 300

The number of students who participated in football from school A = 20% of $100k = 20k$

So, the number of students who participated in hockey from school A = $(20k-20)$

And, the number of students who participated in cricket from school A = $\{300-(20k+20k-20)\} = (320-40k)$

Also,

The total number of students who participated from school B = 250

The number of students who participated in football from school B = 15% of $100k = 15k$

So, the number of students who participated in hockey from school B = $(15k-50)$

And, the number of students who participated in cricket from school B = $\{250-(15k+15k-50)\} = (300-30k)$

Now,

$$(320-40k) + (300-30k) = (135 \times 2)$$

$$620 - 70k = 270$$

$$70k = 350$$

$$k = 5$$

Therefore, the total number of students who participated from all five schools together = $100k = (100 \times 5) = 500$.

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School A

The total number of students who participated = 300

The number of students who participated in football = 20% of 500 = 100

The number of students who participated in hockey = $(100-20) = 80$

The number of students who participated in cricket = $\{300-(100+80)\} = (300-180) = 120$

School B

The total number of students who participated = 250

The number of students who participated in football = 15% of 500 = 75

The number of students who participated in hockey = $(75-50) = 25$

The number of students who participated in cricket = $\{250-(75+25)\} = (250-100) = 150$

School C

The total number of students who participated = 450

The number of students who participated in football = 30% of 500 = 150

The number of students who participated in hockey = $(150-50) = 100$

The number of students who participated in cricket = $\{450-(150+100)\} = (450-250) = 200$

School D

The total number of students who participated = 275

The number of students who participated in football = 25% of 500 = 125

The number of students who participated in hockey = $(125-75) = 50$

The number of students who participated in cricket = $\{275-(125+50)\} = (275-175) = 100$

School E

The total number of students who participated = 150

The number of students who participated in football = 10% of 500 = 50

The number of students who participated in hockey = $(50-20) = 30$

The number of students who participated in cricket = $\{150-(50+30)\} = (150-80) = 70$

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School	The total number of students who participated	The number of students who participated in cricket	The number of students who participated in football	The number of students who participated in hockey
A	300	120	100	80
B	250	150	75	25
C	450	200	150	100
D	275	100	125	50
E	150	70	50	30
Total	1425	640	500	285

Answer: D

School F

The total number of students who participated = 200

The number of students who participated in football = 50

The number of students who participated in hockey = 20% of 200 = 40

Therefore, the number of students who participated in cricket = $\{200 - (50 + 40)\} = (200 - 90) = 110$.

Hence, the correct answer is option D.

9. Questions

Let the total number of students who participated in football from all five schools together be 100k.

Now,

The total number of students who participated from school A = 300

The number of students who participated in football from school A = 20% of 100k = 20k

So, the number of students who participated in hockey from school A = (20k-20)

And, the number of students who participated in cricket from school A = $\{300 - (20k + 20k - 20)\} = (320 - 40k)$

Also,

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The total number of students who participated from school B = 250

The number of students who participated in football from school B = 15% of 100k = 15k

So, the number of students who participated in hockey from school B = (15k-50)

And, the number of students who participated in cricket from school B = $\{250 - (15k + 15k - 50)\} = (300 - 30k)$

Now,

$$(320-40k) + (300-30k) = (135*2)$$

$$620 - 70k = 270$$

$$70k = 350$$

$$k = 5$$

Therefore, the total number of students who participated from all five schools together = $100k = (100*5) = 500$.

School A

The total number of students who participated = 300

The number of students who participated in football = 20% of 500 = 100

The number of students who participated in hockey = $(100-20) = 80$

The number of students who participated in cricket = $\{300-(100+80)\} = (300-180) = 120$

School B

The total number of students who participated = 250

The number of students who participated in football = 15% of 500 = 75

The number of students who participated in hockey = $(75-50) = 25$

The number of students who participated in cricket = $\{250-(75+25)\} = (250-100) = 150$

School C

The total number of students who participated = 450

The number of students who participated in football = 30% of 500 = 150

The number of students who participated in hockey = $(150-50) = 100$

The number of students who participated in cricket = $\{450-(150+100)\} = (450-250) = 200$

School D

The total number of students who participated = 275

The number of students who participated in football = 25% of 500 = 125

The number of students who participated in hockey = $(125-75) = 50$ @QuizByMickey

The number of students who participated in cricket = $\{275-(125+50)\} = (275-175) = 100$

School E

The total number of students who participated = 150

The number of students who participated in football = 10% of 500 = 50

The number of students who participated in hockey = $(50-20) = 30$

The number of students who participated in cricket = $\{150-(50+30)\} = (150-80) = 70$

School	The total number of students who participated	The number of students who participated in cricket	The number of students who participated in football	The number of students who participated in hockey
A	300	120	100	80
B	250	150	75	25
C	450	200	150	100
D	275	100	125	50
E	150	70	50	30
Total	1425	640	500	285

Answer: D

X = The total number of students who participated in football from hockey from school B = $(75+25) = 100$

Y = The difference between the number of students who participated in cricket from school C and E = $(200-70) = 130$

Z = The average number of students who participated in hockey from schools A and C = $\{(80+100)/2\} = (180/2) = 90$

Therefore, $X < Y > Z$.

Hence, the correct answer is option B.

10. Questions

Let the total number of students who participated in football from all five schools together be 100k.

Now,

The total number of students who participated from school A = 300

The number of students who participated in football from school A = 20% of 100k = 20k

So, the number of students who participated in hockey from school A = $(20k-20)$

And, the number of students who participated in cricket from school A = $\{300-(20k+20k-20)\} = (320-40k)$

Also,

The total number of students who participated from school B = 250

The number of students who participated in football from school B = 15% of 100k = 15k

So, the number of students who participated in hockey from school B = $(15k-50)$

And, the number of students who participated in cricket from school B = $\{250-(15k+15k-50)\} = (300-30k)$

Now,

$$(320-40k) + (300-30k) = (135*2)$$

$$620 - 70k = 270$$

$$70k = 350$$

$$k = 5$$

Therefore, the total number of students who participated from all five schools together = $100k = (100*5) = 500$.

School A

The total number of students who participated = 300

The number of students who participated in football = 20% of 500 = 100

The number of students who participated in hockey = $(100-20) = 80$

The number of students who participated in cricket = $\{300-(100+80)\} = (300-180) = 120$

School B

The total number of students who participated = 250

The number of students who participated in football = 15% of 500 = 75

The number of students who participated in hockey = $(75-50) = 25$

The number of students who participated in cricket = $\{250-(75+25)\} = (250-100) = 150$

School C

The total number of students who participated = 450

The number of students who participated in football = 30% of 500 = 150

The number of students who participated in hockey = $(150-50) = 100$

The number of students who participated in cricket = $\{450-(150+100)\} = (450-250) = 200$

School D

The total number of students who participated = 275

The number of students who participated in football = 25% of 500 = 125

The number of students who participated in hockey = $(125-75) = 50$

The number of students who participated in cricket = $\{275-(125+50)\} = (275-175) = 100$

School E

The total number of students who participated = 150

The number of students who participated in football = 10% of 500 = 50

The number of students who participated in hockey = $(50-20) = 30$

The number of students who participated in cricket = $\{150 - (50 + 30)\} = (150 - 80) = 70$

School	The total number of students who participated	The number of students who participated in cricket	The number of students who participated in football	The number of students who participated in hockey
A	300	120	100	80
B	250	150	75	25
C	450	200	150	100
D	275	100	125	50
E	150	70	50	30
Total	1425	640	500	285

Answer: B

Statement I

The ratio of the number of students who participated in cricket and hockey from school E = $70:30 = 7:3$.

So, statement I is not true.

Statement II

The total number of students who participated from school C = 450

And, the number of students who participated in football from school C = 150

The required percentage = $\{(150/450) \times 100\} \% = 33.33\%$.

So, statement II is definitely true.

Therefore, only statement II is definitely true.

Hence, the correct answer is option B.

11. Questions

Answer: A

Series I: 11, 15, 9, 17, __[X]__, 18

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1st term = 11

2nd term = $(11 + 4) = 15$

3rd term = $(15 - 6) = 9$

4th term = $(9 + 8) = 17$

5th term = $(17 - 10) = 7(X)$

$$6^{\text{th}} \text{ term} = (7+12) = 19$$

So, the missing term = $X = 7$

Series II: $[X-6]$, _____, _____, _____, _____, $[Y]$

$[7-6]$, _____, _____, _____, _____, $[Y]$

1, _____, _____, _____, _____, $[Y]$

$$1^{\text{st}} \text{ term} = 1$$

$$2^{\text{nd}} \text{ term} = (1+4) = 5$$

$$3^{\text{rd}} \text{ term} = (5-6) = -1$$

$$4^{\text{th}} \text{ term} = (-1+8) = 7$$

$$5^{\text{th}} \text{ term} = (7-10) = -3$$

$$6^{\text{th}} \text{ term} = (-3+12) = 9 \text{ (Y)}$$

So, the missing term = $Y = 9$

I: Both 7 and 9 are consecutive natural numbers.

So, I is not true.

II: 9 is not a perfect cube.

So, II is definitely not true.

III: 7 is not an odd integer.

So, III is definitely true.

Therefore, only III is definitely true.

Hence, the correct answer is option A.

12. Questions

Answer: E

Series I: 40, 10, 80, 20, 180, 40

$$1^{\text{st}} \text{ term} = 40$$

$$2^{\text{nd}} \text{ term} = (40/4) = 10$$

$$3^{\text{rd}} \text{ term} = (10*8) = 80$$

$$4^{\text{th}} \text{ term} = (80/4) = 20$$

$$5^{\text{th}} \text{ term} = (20*8) = 160 \text{ (180 is wrong)}$$

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$$6^{\text{th}} \text{ term} = (160/4) = 40$$

So, the wrong term = 180

And, the correct term = 160

Series II: __[A]__, 40, 320, 80, 640, __[B]__

160, 40, 320, 80, 640, __[B]__

$$1^{\text{st}} \text{ term} = 160$$

$$2^{\text{nd}} \text{ term} = (160/4) = 40$$

$$3^{\text{rd}} \text{ term} = (40*8) = 320$$

$$4^{\text{th}} \text{ term} = (320/4) = 80$$

$$5^{\text{th}} \text{ term} = (80*8) = 640$$

$$6^{\text{th}} \text{ term} = (640/4) = 160 \text{ (B)}$$

So, A = 160

And, B = 160

Therefore, A = B.

Hence, the correct answer is option E.

13. Questions

Answer: B

Series I: 24, 13, 15, 24.50, 55, 142.50

$$1^{\text{st}} \text{ term} = 24$$

$$2^{\text{nd}} \text{ term} = (24*0.50+1) = (12+1) = 13$$

$$3^{\text{rd}} \text{ term} = (13*1+2) = (13+2) = 15$$

$$4^{\text{th}} \text{ term} = (15*1.50+3) = (22.50+3) = 25.50 \text{ (24.50 is wrong)}$$

$$5^{\text{th}} \text{ term} = (25.50*2+4) = (51+4) = 55$$

$$6^{\text{th}} \text{ term} = (55*2.50+5) = (137.50+5) = 142.50$$

So, the wrong term = 24.50

And, the correct term = 25.50

So, m = 24.50

Series II: 30, 41, 54, 71, 91, 113

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1st term = 30

2nd term = $(30+11) = 41$

3rd term = $(41+13) = 54$

4th term = $(54+17) = 71$

5th term = $(71+19) = 90$ (91 is wrong)

6th term = $(90+23) = 113$

So, the wrong term = 91

And, the correct term = 90

So, $n = 91$

Therefore, the required value = $\{(n+7)/2m\} = \{(91+7)/(2*24.50)\} = (98/49) = 2$.

Hence, the correct answer is option B.

14. Questions

Answer: C

The possible values of $R = 1, 4$ or 9

1 is a perfect square of 1, which is not a prime number.

So, R cannot be 1.

4 is a perfect square of 2, which is a prime number but not odd.

So, R cannot be 4.

9 is a perfect square of 3, which is an odd prime number.

So, $R = 9$ (confirmed)

$P + Q = R$

$P + Q = 9$

P is a composite number.

The possible values of $P = 4, 6$, or 8 .

If $P = 4$ then $Q = 5$ ($Q > P$).

This is satisfied.

If $P = 6$ or 8 then $Q = 3$ or 1 .

This is not satisfied.

Now, the only possible values of P and $Q = 4$ and 5 ($Q > P$)

So, $P = 4$

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And, $Q = 5$

Series: [P], [Q], [R], __[?], 292, 3417

4, 5, 9, __[?], 292, 3417

1st term = 4

2nd term = $(4+1^1) = (4+1) = 5$

3rd term = $(5+2^2) = (5+4) = 9$

4th term = $(9+3^3) = (9+27) = 36$ (?)

5th term = $(36+4^4) = (36+256) = 292$

6th term = $(292+5^5) = (292+3125) = 3417$

Therefore, ? = 36

Hence, the correct answer is option C.

15. Questions

Answer: D

Series: 23, 20, 26, 14, 38, -10

1st term = 23

2nd term = $(23-3) = 20$

3rd term = $(20+6) = 26$

4th term = $(26-12) = 14$

5th term = $(14+24) = 38$

6th term = $(38-48) = -10$

Another Series

1st term = $4^2 = 16$ (4 is the smallest composite number)

2nd term = $(16-3) = 13$

3rd term = $(13+6) = 19$

4th term = $(19-12) = 7$

5th term = $(7+24) = 31$

6th term = $(31-48) = -17$

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Therefore, the sixth term of another series = -17.

Hence, the correct answer is option D.

16. Questions

The number of overs bowled on each day,

$$\text{Day 1} = 80$$

$$\text{Day 2} = (200-80) = 120$$

$$\text{Day 3} = (240-200) = 40$$

$$\text{Day 4} = (300-240) = 60$$

$$\text{Day 5} = (400-300) = 100$$

So, the number of balls faced on each day,

$$\text{Day 1} = (80 \times 6) = 480$$

$$\text{Day 2} = (120 \times 6) = 720$$

$$\text{Day 3} = (40 \times 6) = 240$$

$$\text{Day 4} = (60 \times 6) = 360$$

$$\text{Day 5} = (100 \times 6) = 600$$

Now,

The ratio of the number of runs scored on days 4 and 5 = $45:60 = 3:4$. Let it be $3k$ and $4k$ respectively.

$$\text{So, the strike-rate on day 4} = \{(3k/360) \times 100\} = (5k/6)$$

$$\text{And, the strike-rate on day 5} = \{(4k/600) \times 100\} = (2k/3)$$

Now,

$$(5k/6) - (2k/3) = 30$$

$$(5k-4k)/6 = 30$$

$$k = 180$$

$$\text{So, the number of runs scored on day 4} = 3k = (3 \times 180) = 540$$

$$\text{And, the number of runs scored on day 5} = 4k = (4 \times 180) = 720$$

Day 1

$$\text{The number of runs scored by A} = \{(540/45) \times 30\} = 360$$

$$\text{The number of balls faced by A} = 480$$

$$\text{The strike-rate of A} = \{(360/480) \times 100\} = 75$$

Day 2

$$\text{The number of runs scored by A} = \{(540/45) \times 48\} = 576$$

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The number of balls faced by A = 720

The strike-rate of A = $\{(576/720)*100\} = 80$

Day 3

The number of runs scored by A = $\{(540/45)*20\} = 240$

The number of balls faced by A = 240

The strike-rate of A = $\{(240/240)*100\} = 100$

Day 4

The number of runs scored by A = 540

The number of balls faced by A = 360

The strike-rate of A = $\{(540/360)*100\} = 150$

Day 5

The number of runs scored by A = 720

The number of balls faced by A = 600

The strike-rate of A = $\{(720/600)*100\} = 120$

Day	The number of runs scored	The number of balls faced	The strike-rate
1	360	480	75
2	576	720	80
3	240	240	100
4	540	360	150
5	720	600	120

Answer: C

Now, according to the question,

Day 1

The total number of runs scored by A = 360

Let the number of fours and sixes scored be $6n$ and n respectively.

So, the number of runs scored in fours = $(4*6n) = 24n$

And, the number of runs scored in sixes = $(6*n) = 6n$

The total number of runs scored in boundaries (fours + sixes) = $(24n+6n) = 30n$

So, the number of runs scored in other than boundaries = $(360-30n)$

Now,

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$$24n/(360-30n) = 16/100$$

$$24n/(360-30n) = 4/25$$

$$600n = 1440 - 120n$$

$$720n = 1440$$

$$n = 2$$

Therefore, the total number of runs scored in boundaries (fours + sixes) = $30n = (30 \times 2) = 60$.

Hence, the correct answer is option C.

17. Questions

The number of overs bowled on each day,

$$\text{Day 1} = 80$$

$$\text{Day 2} = (200-80) = 120$$

$$\text{Day 3} = (240-200) = 40$$

$$\text{Day 4} = (300-240) = 60$$

$$\text{Day 5} = (400-300) = 100$$

So, the number of balls faced on each day,

$$\text{Day 1} = (80 \times 6) = 480$$

$$\text{Day 2} = (120 \times 6) = 720$$

$$\text{Day 3} = (40 \times 6) = 240$$

$$\text{Day 4} = (60 \times 6) = 360$$

$$\text{Day 5} = (100 \times 6) = 600$$

Now,

The ratio of the number of runs scored on days 4 and 5 = $45:60 = 3:4$. Let it be $3k$ and $4k$ respectively.

$$\text{So, the strike-rate on day 4} = \{(3k/360) \times 100\} = (5k/6)$$

$$\text{And, the strike-rate on day 5} = \{(4k/600) \times 100\} = (2k/3)$$

Now,

$$(5k/6) - (2k/3) = 30$$

$$(5k-4k)/6 = 30$$

$$k = 180$$

$$\text{So, the number of runs scored on day 4} = 3k = (3 \times 180) = 540$$

$$\text{And, the number of runs scored on day 5} = 4k = (4 \times 180) = 720$$

Day 1

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The number of runs scored by A = $\{(540/45)*30\} = 360$

The number of balls faced by A = 480

The strike-rate of A = $\{(360/480)*100\} = 75$

Day 2

The number of runs scored by A = $\{(540/45)*48\} = 576$

The number of balls faced by A = 720

The strike-rate of A = $\{(576/720)*100\} = 80$

Day 3

The number of runs scored by A = $\{(540/45)*20\} = 240$

The number of balls faced by A = 240

The strike-rate of A = $\{(240/240)*100\} = 100$

Day 4

The number of runs scored by A = 540

The number of balls faced by A = 360

The strike-rate of A = $\{(540/360)*100\} = 150$

Day 5

The number of runs scored by A = 720

The number of balls faced by A = 600

The strike-rate of A = $\{(720/600)*100\} = 120$

Day	The number of runs scored	The number of balls faced	The strike-rate
1	360	480	75
2	576	720	80
3	240	240	100
4	540	360	150
5	720	600	120

Answer: D

Day 2

The number of runs scored = 576

The number of balls faced (excluding dot balls) = $(720-2X)$

So, the strike-rate = $[\{576/(720-2X)\} \times 100]$

Now,

$$[\{576/(720-2X)\} * 100] = 120\% \text{ of } 80$$

$$576(720-2X) = 96/100$$

$$576(720-2X) = 24/25$$

$$17280 - 48X = 14400$$

$$48X = 2880$$

$$X = 60$$

$$\text{Therefore, the required value} = \sqrt{(X-11)} = \sqrt{(60-11)} = \sqrt{49} = 7.$$

Hence, the correct answer is option D.

18. Questions

The number of overs bowled on each day,

$$\text{Day 1} = 80$$

$$\text{Day 2} = (200-80) = 120$$

$$\text{Day 3} = (240-200) = 40$$

$$\text{Day 4} = (300-240) = 60$$

$$\text{Day 5} = (400-300) = 100$$

So, the number of balls faced on each day,

$$\text{Day 1} = (80 \times 6) = 480$$

$$\text{Day 2} = (120 \times 6) = 720$$

$$\text{Day 3} = (40 \times 6) = 240$$

$$\text{Day 4} = (60 \times 6) = 360$$

$$\text{Day 5} = (100 \times 6) = 600$$

Now,

The ratio of the number of runs scored on days 4 and 5 = 45:60 = 3:4. Let it be 3k and 4k respectively.

$$\text{So, the strike-rate on day 4} = \{(3k/360) * 100\} = (5k/6)$$

$$\text{And, the strike-rate on day 5} = \{(4k/600) * 100\} = (2k/3)$$

Now,

$$(5k/6) - (2k/3) = 30$$

$$(5k-4k)/6 = 30$$

$$k = 180$$

$$\text{So, the number of runs scored on day 4} = 3k = (3 * 180) = 540$$

And, the number of runs scored on day 5 = $4k = (4 \times 180) = 720$

Day 1

The number of runs scored by A = $\{(540/45) \times 30\} = 360$

The number of balls faced by A = 480

The strike-rate of A = $\{(360/480) \times 100\} = 75$

Day 2

The number of runs scored by A = $\{(540/45) \times 48\} = 576$

The number of balls faced by A = 720

The strike-rate of A = $\{(576/720) \times 100\} = 80$

Day 3

The number of runs scored by A = $\{(540/45) \times 20\} = 240$

The number of balls faced by A = 240

The strike-rate of A = $\{(240/240) \times 100\} = 100$

Day 4

The number of runs scored by A = 540

The number of balls faced by A = 360

The strike-rate of A = $\{(540/360) \times 100\} = 150$

Day 5

The number of runs scored by A = 720

The number of balls faced by A = 600

The strike-rate of A = $\{(720/600) \times 100\} = 120$

Day	The number of runs scored	The number of balls faced	The strike-rate
1	360	480	75
2	576	720	80
3	240	240	100
4	540	360	150
5	720	600	120

Answer: B

Day 3

The strike-rate on day 3 = 100

The number of overs faced by A = 50

So, the number of balls faced by A = $(50 \times 6) = 300$

Let the number of runs scored be m.

So,

$$(m/300) \times 100 = 100$$

$$m/300 = 100/100$$

$$m/300 = 1/1$$

$$m = 300$$

Now, the number of runs scored by A = m = 300

And, the number of runs scored by A initially = 240

Therefore, the required percentage increase in the number of runs scored = $\left[\frac{(300-240)}{240} \times 100\right]\% = \left[\frac{60}{240} \times 100\right]\% = 25\%$.

Hence, the correct answer is option B.

19. Questions

The number of overs bowled on each day,

$$\text{Day 1} = 80$$

$$\text{Day 2} = (200-80) = 120$$

$$\text{Day 3} = (240-200) = 40$$

$$\text{Day 4} = (300-240) = 60$$

$$\text{Day 5} = (400-300) = 100$$

So, the number of balls faced on each day,

$$\text{Day 1} = (80 \times 6) = 480$$

$$\text{Day 2} = (120 \times 6) = 720$$

$$\text{Day 3} = (40 \times 6) = 240$$

$$\text{Day 4} = (60 \times 6) = 360$$

$$\text{Day 5} = (100 \times 6) = 600$$

Now,

The ratio of the number of runs scored on days 4 and 5 = $45:60 = 3:4$. Let it be 3k and 4k respectively.

So, the strike-rate on day 4 = $\left\{\frac{(3k/360) \times 100}{1}\right\} = (5k/6)$

And, the strike-rate on day 5 = $\left\{\frac{(4k/600) \times 100}{1}\right\} = (2k/3)$

Now,

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$$(5k/6) - (2k/3) = 30$$

$$(5k-4k)/6 = 30$$

$$k = 180$$

So, the number of runs scored on day 4 = $3k = (3 \times 180) = 540$

And, the number of runs scored on day 5 = $4k = (4 \times 180) = 720$

Day 1

The number of runs scored by A = $\{(540/45) \times 30\} = 360$

The number of balls faced by A = 480

The strike-rate of A = $\{(360/480) \times 100\} = 75$

Day 2

The number of runs scored by A = $\{(540/45) \times 48\} = 576$

The number of balls faced by A = 720

The strike-rate of A = $\{(576/720) \times 100\} = 80$

Day 3

The number of runs scored by A = $\{(540/45) \times 20\} = 240$

The number of balls faced by A = 240

The strike-rate of A = $\{(240/240) \times 100\} = 100$

Day 4

The number of runs scored by A = 540

The number of balls faced by A = 360

The strike-rate of A = $\{(540/360) \times 100\} = 150$

Day 5

The number of runs scored by A = 720

The number of balls faced by A = 600

The strike-rate of A = $\{(720/600) \times 100\} = 120$

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Day	The number of runs scored	The number of balls faced	The strike-rate
1	360	480	75
2	576	720	80
3	240	240	100
4	540	360	150
5	720	600	120

Answer: A

The total number of overs faced by A till day 5 = 400

And, the total overs faced by A on 6 days together = $(90 \times 6) = 540$

So, the number of overs faced by A on day 6 = $(540 - 400) = 140$

Therefore, the number of balls faced by A on day 6 = $(140 \times 6) = 840$

Let the number of runs scored on day 6 be a.

So, the strike-rate on day 6 = $\{(a/840) \times 100\} = (5a/42)$

Now,

$$(5a/42) = 50$$

$$5a = 2100$$

$$a = 420$$

So, the number of runs scored on day 6 = $a = 420$

Therefore, the run rate on day 6 = $(420/140) = 3$.

Hence, the correct answer is option A.

20. Questions

The number of overs bowled on each day,

$$\text{Day 1} = 80$$

$$\text{Day 2} = (200 - 80) = 120$$

$$\text{Day 3} = (240 - 200) = 40$$

$$\text{Day 4} = (300 - 240) = 60$$

$$\text{Day 5} = (400 - 300) = 100$$

So, the number of balls faced on each day,

$$\text{Day 1} = (80 \times 6) = 480$$

$$\text{Day 2} = (120 \times 6) = 720$$

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$$\text{Day 3} = (40 \times 6) = 240$$

$$\text{Day 4} = (60 \times 6) = 360$$

$$\text{Day 5} = (100 \times 6) = 600$$

Now,

The ratio of the number of runs scored on days 4 and 5 = $45:60 = 3:4$. Let it be $3k$ and $4k$ respectively.

$$\text{So, the strike-rate on day 4} = \{(3k/360) \times 100\} = (5k/6)$$

$$\text{And, the strike-rate on day 5} = \{(4k/600) \times 100\} = (2k/3)$$

Now,

$$(5k/6) - (2k/3) = 30$$

$$(5k - 4k)/6 = 30$$

$$k = 180$$

$$\text{So, the number of runs scored on day 4} = 3k = (3 \times 180) = 540$$

$$\text{And, the number of runs scored on day 5} = 4k = (4 \times 180) = 720$$

Day 1

$$\text{The number of runs scored by A} = \{(540/45) \times 30\} = 360$$

$$\text{The number of balls faced by A} = 480$$

$$\text{The strike-rate of A} = \{(360/480) \times 100\} = 75$$

Day 2

$$\text{The number of runs scored by A} = \{(540/45) \times 48\} = 576$$

$$\text{The number of balls faced by A} = 720$$

$$\text{The strike-rate of A} = \{(576/720) \times 100\} = 80$$

Day 3

$$\text{The number of runs scored by A} = \{(540/45) \times 20\} = 240$$

$$\text{The number of balls faced by A} = 240$$

$$\text{The strike-rate of A} = \{(240/240) \times 100\} = 100$$

Day 4

$$\text{The number of runs scored by A} = 540$$

$$\text{The number of balls faced by A} = 360$$

$$\text{The strike-rate of A} = \{(540/360) \times 100\} = 150$$

Day 5

$$\text{The number of runs scored by A} = 720$$

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The number of balls faced by A = 600

The strike-rate of A = $\{(720/600)*100\} = 120$

Day	The number of runs scored	The number of balls faced	The strike-rate
1	360	480	75
2	576	720	80
3	240	240	100
4	540	360	150
5	720	600	120

Answer: B

The total number of runs scored by A on day 4 = 540

So, the number of runs scored in boundaries by A on day 4 = $1/9$ of 540 = 60

And, the number of runs scored other than boundaries on day 4 = $(540-60) = 480$

Also,

The total number of runs scored by A on day 5 = 720

The number of runs scored other than boundaries on day 5 = $(480+150) = 630$

So, the number of runs scored in boundaries on day 5 = $(720-630) = 90$

Therefore, the required percentage = $(90/720)*100 = 12.50\%$.

Hence, the correct answer is option B.

@QuizByMickey